Understanding Web Applications

Lesson 4

Objective Domain Matrix

Skills/Concepts	MTA Exam Objectives
Understanding Web Page Development	Understand Web page development (4.1)
Understanding ASP.NET Application Development	Understand Microsoft ASP.NET Web Application development (4.2)
Understanding IIS Web Hosting	Understand Web hosting (4.3)
Understanding Web Services Development	Understand Web Services (4.4)

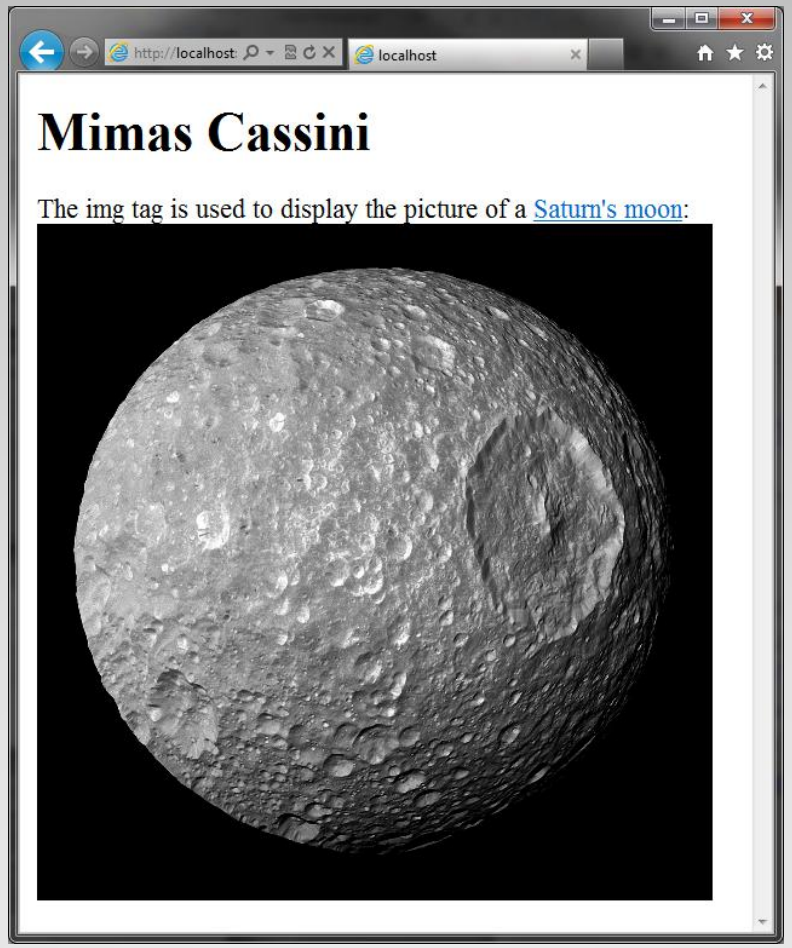
HTML

- Hypertext Markup Language (HTML) is the language used by Web servers and browsers to describe a Web page.
- An HTML page has two distinct parts: a header and a body.
- HTML tags define the structure and content of a page. Each starting tag has a matching ending tag. For example, the ending tag for <html> is </html>.
- The header is enclosed within the <head> and </head> tags.
- The body is enclosed within the <body> and </body> tags

HTML Example

```
<html xmlns="http://www.w3.org/1999/xhtml">
<head>
    <title></title>
</head>
<body>
<h1>Mimas Cassini</h1>
The img tag is used to display the picture
of a <a href="http://en.wikipedia.org/wiki/Mimas_(moon)">
Saturn's moon</a>: <br />
<img  height="400px" width="400px" alt="Mimas Cassini" src=
"http://upload.wikimedia.org/wikipedia/commons/b/bc/Mimas_Cassini.jpg"/>
</body>
</html>
```

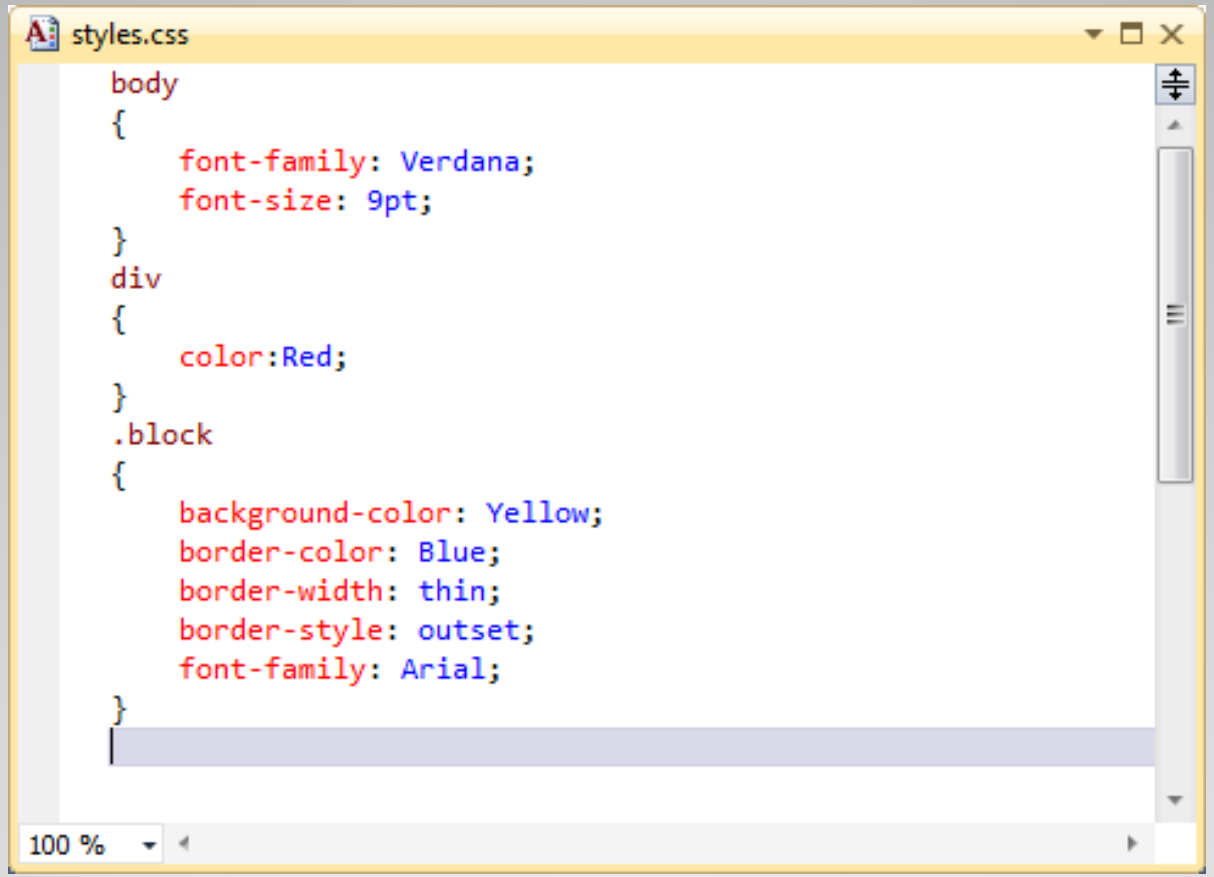
HTML Output


http://localhost
localhost
Mimas Cassini
The img tag is used to display the picture of a Saturn's moon:

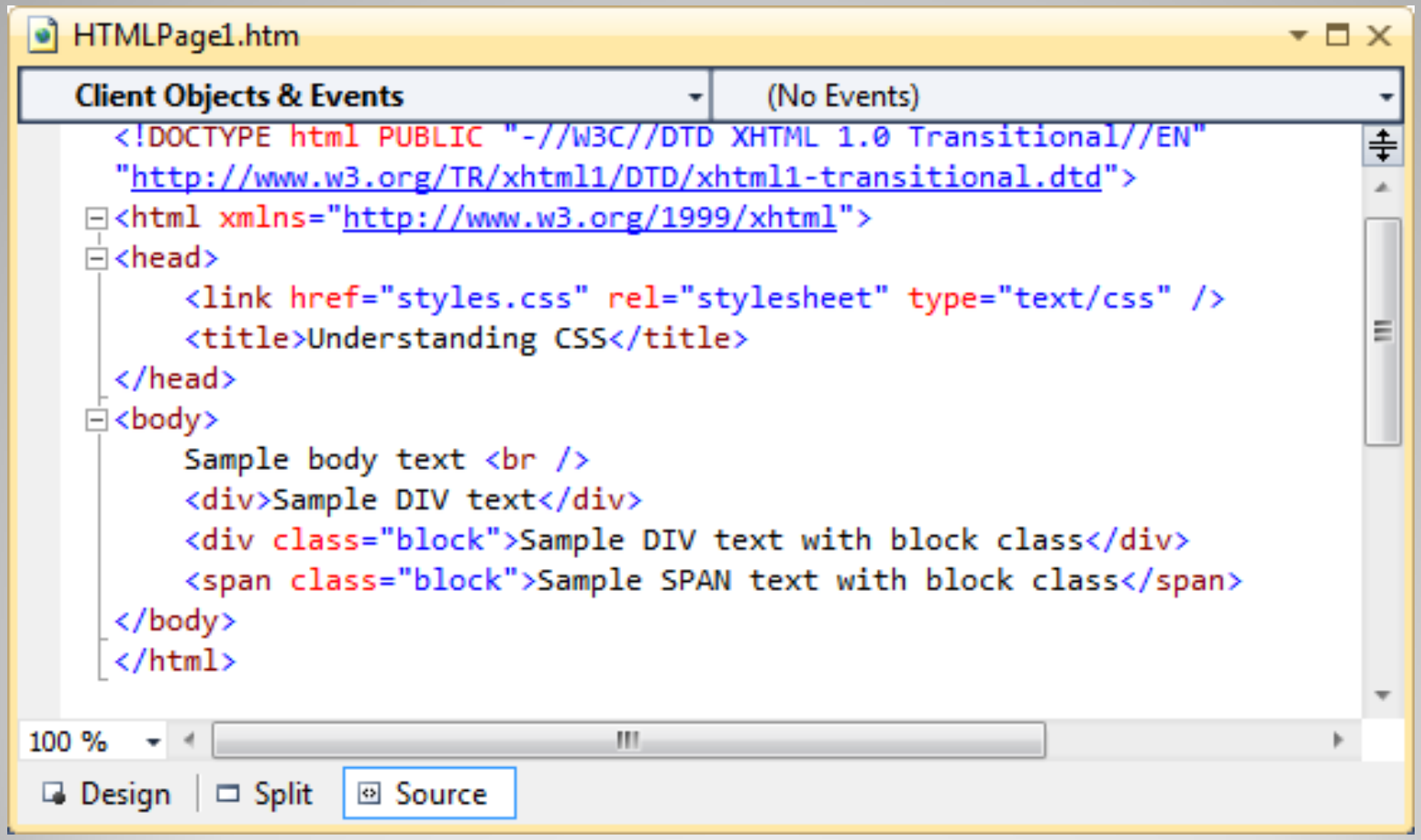
Cascading Style Sheets

- Cascading style sheets (CSS) is a language that describes information about displaying a Web page.
- CSS enable you to store a Web page's style and formatting information separate from the HTML code.
- HTML specifies what will be displayed whereas CSS specifies how that material will be displayed.
- When used effectively, CSS is a great tool for increasing site-wide consistency and maintainability.

CSS Example


styles.css
body
{
font-family: Verdana;
font-size: 9pt;
}
div
{
color:Red;
}
.block
{
background-color: Yellow;
border-color: Blue;
border-width: thin;
border-style: outset;
font-family: Arial;
}
100 %

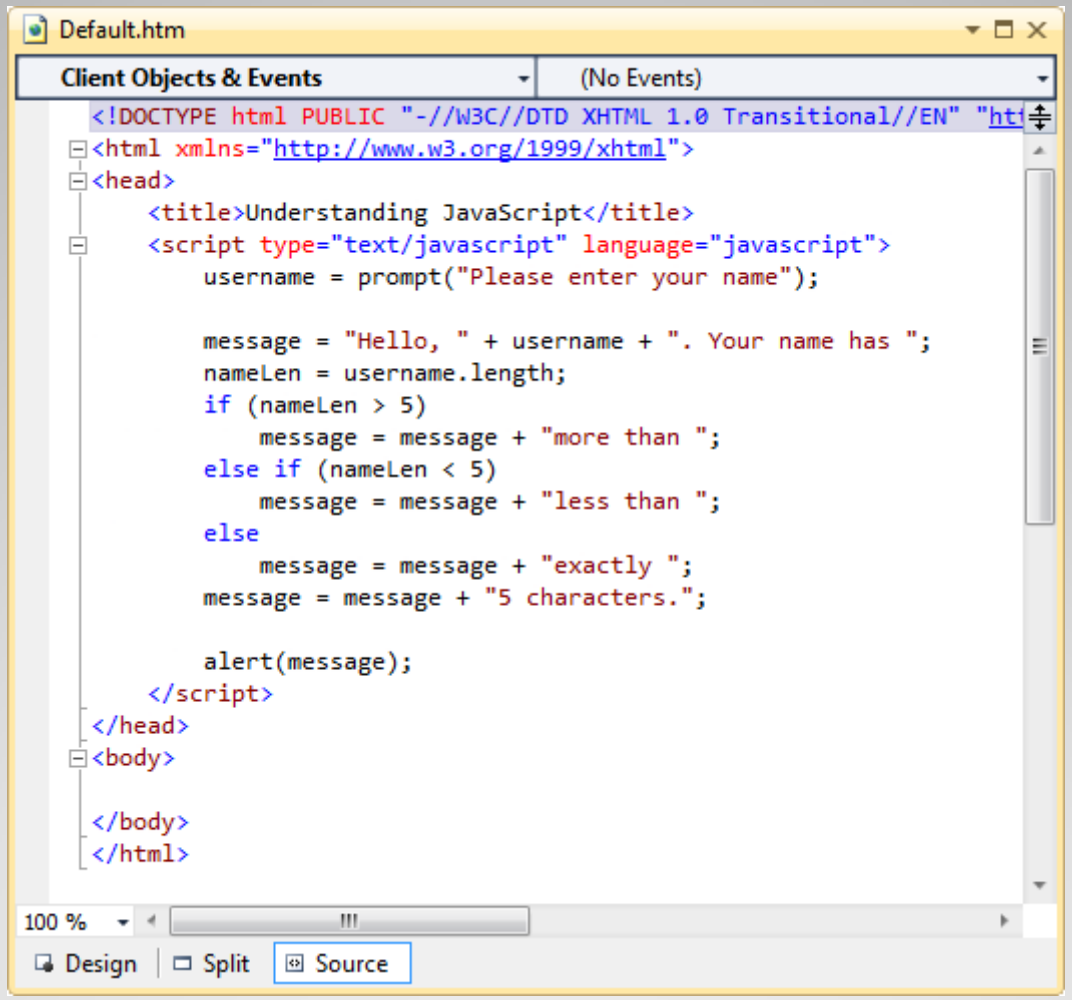
Using the CSS File



JavaScript

- JavaScript is a client-side scripting language.
- JavaScript runs inside Web browsers. JavaScript is supported by all popular Web Browsers.
- Ajax is shorthand for “Asynchronous JavaScript and XML.” Ajax uses JavaScript extensively in order to provide responsive Web applications.
- The ASP.NET AJAX framework lets you implement Ajax functionality on ASP.NET Web pages.

JavaScript Example



Client-Side vs. Server-Side Programming

- Client-side programming refers to programs that execute completely on a user's local computer.
  - Examples: Windows Forms application, JavaScript code that executes within a Web browser.
- Server-side programming refers to programs that are executed completely on a server and make use of the server's computational resources.
  - Examples: Web applications and Web services.
- Hybrid applications use both client- and server-side programming. Ajax applications use a mix of server-side programming and client-side code to create interactive and highly responsive Web applications.

ASP.NET

- ASP.NET is the part of the .NET Framework that enables you to develop Web applications and Web services.

- The ASP.NET infrastructure has two main parts:
  - A set of classes and interfaces that enables communication between the Web browser and Web server. These classes are organized in the System.Web namespace.
  - A runtime process, also known as the ASP.NET worker process (aspnet_wp.exe), that handles the Web request for ASP.NET resources.

ASP.NET Page Execution

- The ASP.NET worker process (aspnet_wp.exe) fulfills the request for ASP.NET page execution.
- The ASP.NET worker process compiles the .aspx file into an assembly and instructs the Common Language Runtime (CLR) to execute the assembly.
- When the assembly executes, it takes the services of various classes in the .NET Framework class library to accomplish its work and generate response messages for the requesting client.
- The ASP.NET worker process collects the responses generated by the execution of the Web page and creates a response packet.

ASP.NET Page Life Cycle

Event	Description
PreInit	Several page properties, such as Request, Response, IsPostBack, and UICulture, are set at this stage.
Init	During the initialization stage, all the controls on the page are initialized and made available.
Load	If the request is a postback, the load stage is used to restore control properties with information from view state and control state.
PreRender	This stage signals that the page is just about to render its contents.
Unload	During the unload stage, the response is sent to the client and page cleanup is performed.

State Management

- The state of a Web page is made up of the values of the various variables and controls.
- State management is the process of preserving the state of a Web page across multiple trips between browser and server.
- State Management Techniques:
  - Client-side state management
  - Server-side state management

Client-Side State Management

- Client-side techniques use HTML code and the capabilities of the Web browser to store state information on the client computer.

- Common client-side state management techniques:
  - Query Strings
  - Cookies
  - Hidden Fields
  - View State

Query Strings

- Query strings stores the data values in the query string portion of a page URL. For example, the following URL embeds a key (“q”) and its value (“television”) in query string: http://www.bing.com/search?q=television.

- To retrieve the value of the key in an ASP.NET page, use the expression:

  Request.QueryString["q"].

- **QueryString** is a property of the Request object, and it gets the collection of all the query-string variables and their values.

Cookies

Cookies are small packets of information that are stored by a Web browser locally on the user's computer. Cookies are commonly used for storing user preferences.

To add a Cookie:

```
HttpCookie cookie = new HttpCookie("Name", "Bob");
cookie.Expires = DateTime.Now.AddMinutes(10);
Response.Cookies.Add(cookie);
```

To read a cookie:

```
if (Request.Cookies["Name"] != null)
{
    name = Request.Cookies["Name"].Value;
}
```

Hidden Fields

- Hidden fields contain information that is not displayed on a Web page but is still part of the page's HTML code.
- Hidden fields can be created by using the following HTML element:

```
<input type="hidden">
```

- The ASP.NET HTML Server control **HtmlInputHidden** also maps to this HTML element.

View State

- ASP.NET uses View State to maintain the state of controls across page postbacks.
- When ASP.NET executes a page, it collects the values of all nonpostback controls that are modified in the code and formats them into a single encoded string. This string is stored in a hidden field in a control named **__VIEWSTATE**.
- View State may increase the size of your page.
- View State is enabled by default by you can disable it either at the control level or at the page level.

Server-Side State Management

- Server-side state management uses server resources to store state information.
- Storing and processing session information on a server increases the server's load and requires additional server resources to serve the Web pages.
- ASP.NET supports server-side state management at two levels:
  - Session State
  - Application State

Session State

- An ASP.NET application creates a unique session for each user who sends a request to the application.
- Session state can be used for temporarily store user data such as shopping cart contents.

- Reading from session:

```
if (Session["Name"] != null)
{
   /* additional code here */
}
```

- Writing to session:

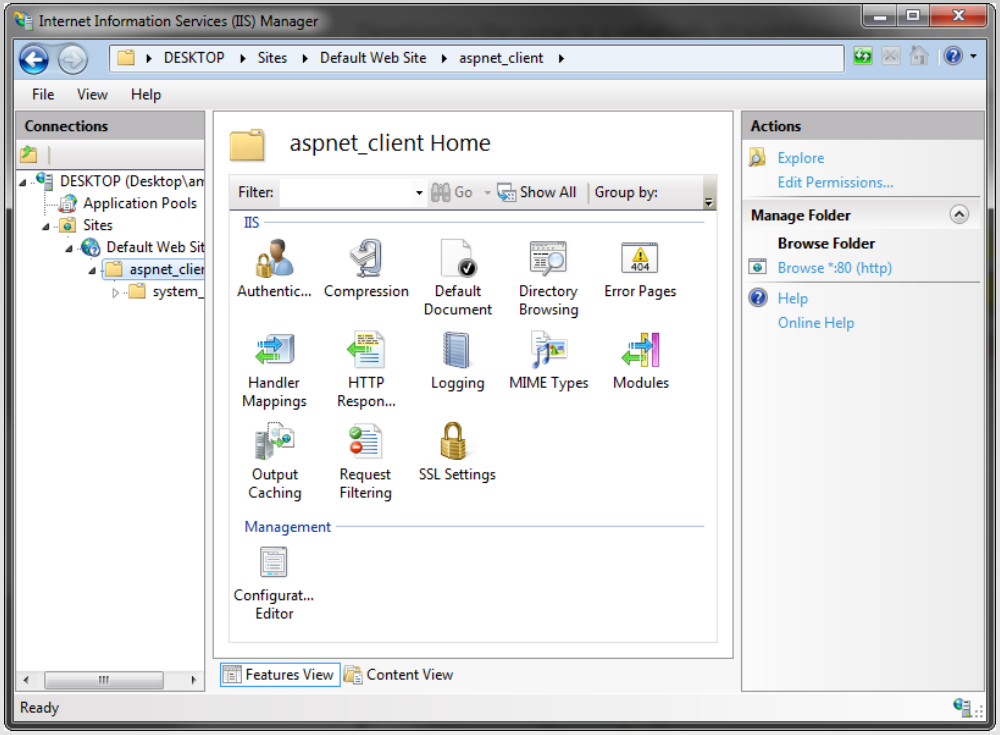
```
Session.Add("Name", TextBox1.Text);
```

Application State

- Application state is used to store data that is used throughout an application.
- Application State is not user-specific.
- Application state can be accessed through the Application property of the Page class.
- The Application property provides access to the **HttpApplicationState** object.
- The **HttpApplicationState** object stores the application state as a collection of key-value pairs.

Internet Information Services

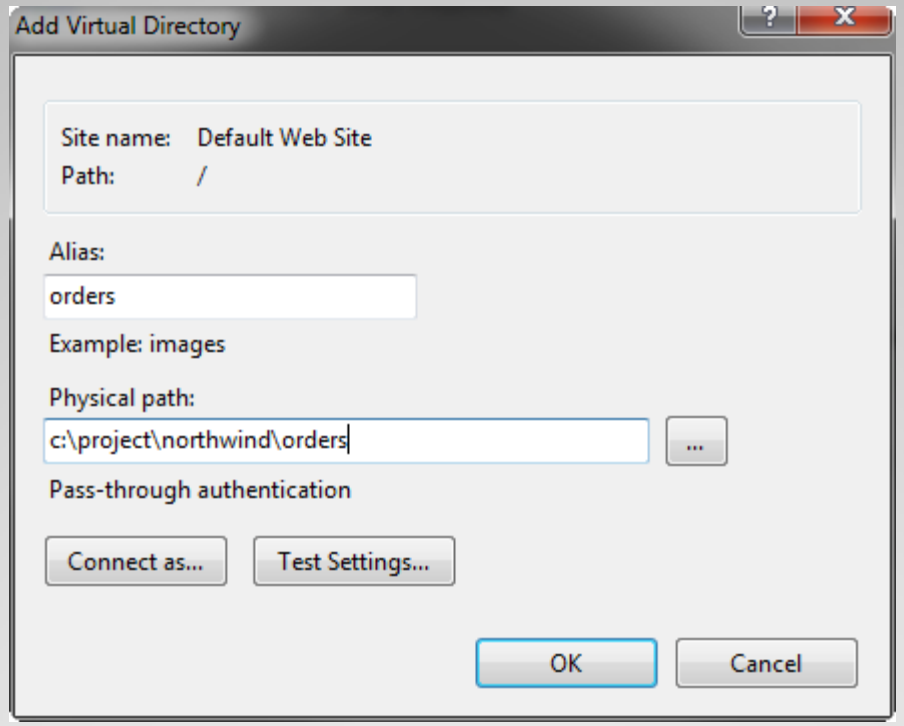
- Internet Information Services (IIS) is a Web server for hosting Web applications on the Windows operating system.



Web Site & Virtual Directories

- An IIS server uses the concepts of sites, applications, and virtual directories.
- A Web site (for example: www.northwind.com) is a container of applications and virtual directories.
- A virtual directory is an alias that maps to a physical directory on the Web server. For example, in the address www.northwind.com/orders, “orders” is a virtual directory.
- A virtual directory maps to a physical directory (for example, c:\inetpub\wwwroot\northwind\orders).

Creating a Virtual Directory



Deploying Web Applications

- Xcopy or FTP

For simple websites that require simply copying the files.

- Windows Installer

For complex websites that require custom actions during the deployment process. Windows Installer can create virtual directories, restart services, register components, etc.

Web Services

- Web services provide a way to interact with programming objects located on remote computers.
- Web services are based on standard technologies and are interoperable.

Key technologies:

- Hypertext Transmission Protocol (HTTP)
- Extensible Markup Language (XML)
- Simple Object Access Protocol (SOAP)
- Web Services Description Language (WSDL)

SOAP

- SOAP is the protocol that defines how remote computers exchange messages as part of a Web service communication.
- Message format: XML
  - XML is easier for otherwise non compatible systems to understand.
- Message transmission: HTTP
  - HTTP, they can normally reach any machine on the Internet without being blocked by firewalls.

WSDL

- WSDL stands for Web Services Description Language.
- A WSDL file acts as the public interface of a Web service and includes the following information:
  - The data types it can process
  - The methods it exposes
  - The URLs through which those methods can be accessed

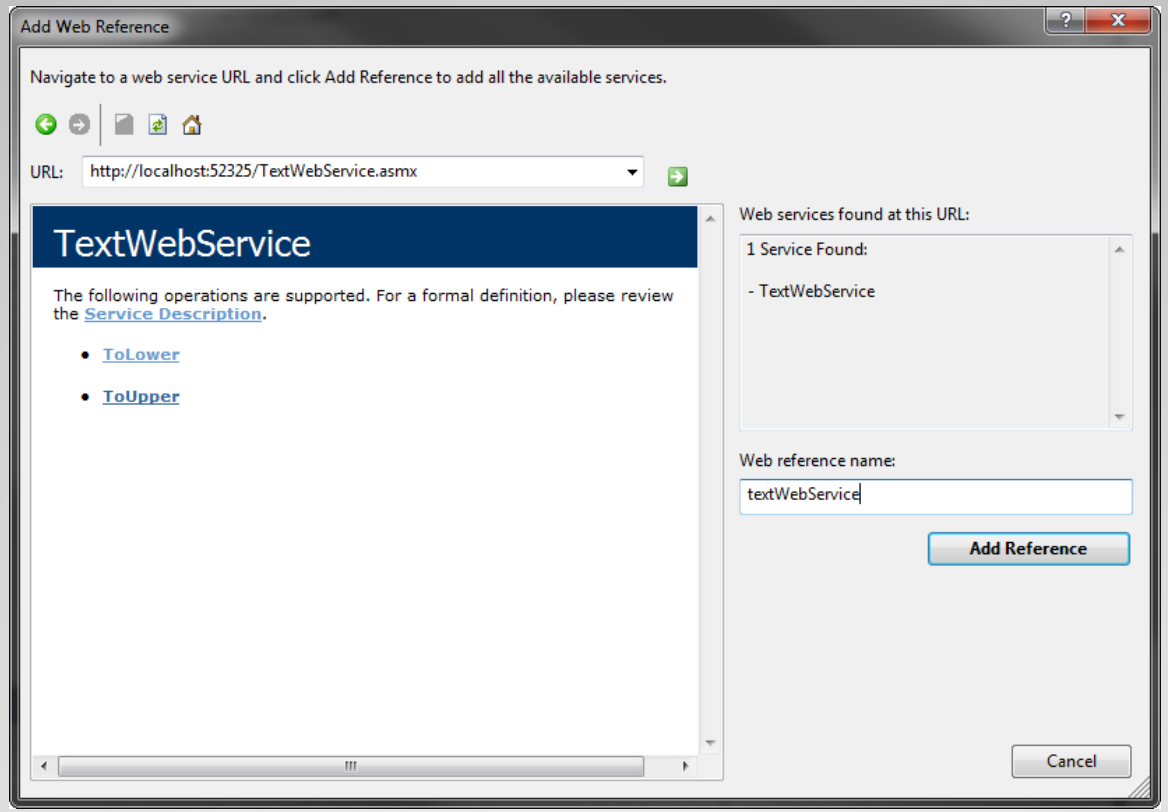
Creating Web Services

- Inherit from System.Web.Services.WebService
- Mark the class with a **WebService** attribute
- Mark the methods with **WebMethod** attribute

```
[WebService(Namespace = "http://northwindtraders.com/")]
[WebServiceBinding(ConformsTo= WsiProfiles.BasicProfile1_1)]
public class TextWebService : System.Web.Services.WebService
{
    [WebMethod]
    public string ToUpper(string inputString)
    {
        return inputString.ToUpper();
    }
}
```

Consuming Web Services

- Add a reference to the Web service by using the Add Web Reference dialog Box.



Consuming Web Services

- The proxy object allows you to invoke Web service methods.

```
protected void Button1_Click(object sender, EventArgs e)
{
    var webService = new textWebService.TextWebService();
    toLowerLabel.Text = webService.ToLower(TextBox1.Text);
    toUpperLabel.Text = webService.ToUpper(TextBox1.Text);
}
```

Recap

- Web Page Development
  - HTML, CSS, JavaScript
- Client-side vs. server-side programming
- ASP.NET Page Life Cycle
  - PreInit, Init, Load, PreRender, and Unload
- State Management
  - Query strings, cookies, hidden fields, viewstate
  - Session state, application state
- IIS Web Hosting
  - Web Sites, Virtual directories
- Web Services
  - SOAP, WSDL
  - WebService attribute, WebMethod attribute